# Applied ML

23-02-26

## Step 4 - Data preprocessing

### 1) Null Values handling

Most of the ML models don't accept null values as input

### Data prunning

eg. data

| area | rooms | price | city |
|---|---|---|---|
| ⋮ | ⋮ | ⋮ | ⋮ |

```
df = pd.read_csv(filepath)

df.isna().sum() / len(df)
```

area → 2% → drop these rows
rooms → 0%
price → 1% → drop these rows
city → 30% → drop this column

} data prunning

area rooms price

build the ML model only on remaining columns

# Data Imputation

- replace null values with valid values using domain specific assumptions

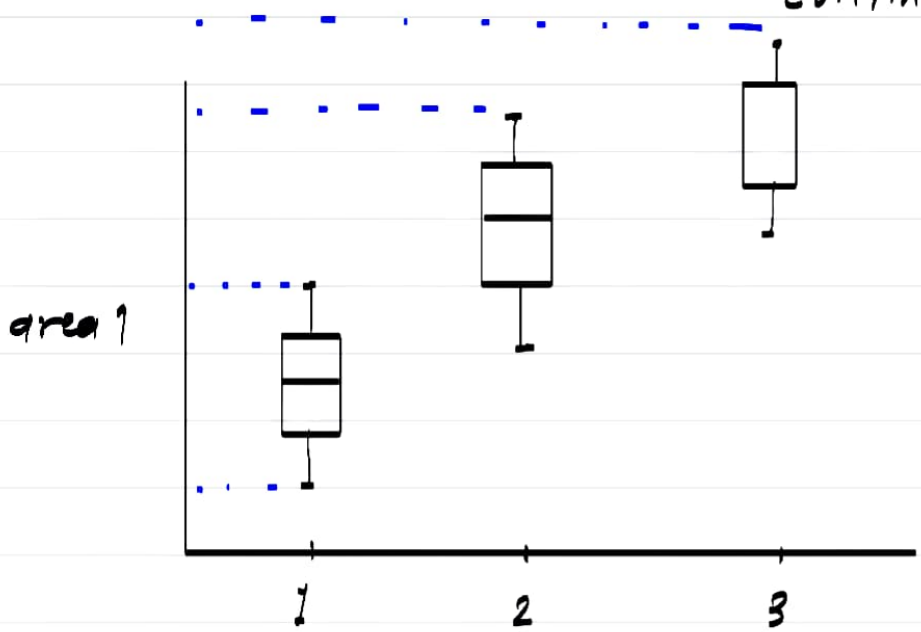
eg.

| area | rooms | price |
|---|---|---|
| 190 | 3 | 10 |
| 195 | null | 12 |
| 185 | 3 | 9 |
| 170 | 2 | 7 |
| 150 | 2 | 8 |
| 110 | 1 | 7 |
| 150 | null | 6 |
| 120 | 1 | 5 |

| area | rooms |
|---|---|
| 100-120 | 1 |
| 120-150 | 2 |
| 150-200 | 3 |

bivariate analysis of area and rooms

↓ continuous → discrete



- Quality of data imputation depends on accuracy of assumption.

- If we are not very much confident on the assumption, then we should not do data imputation

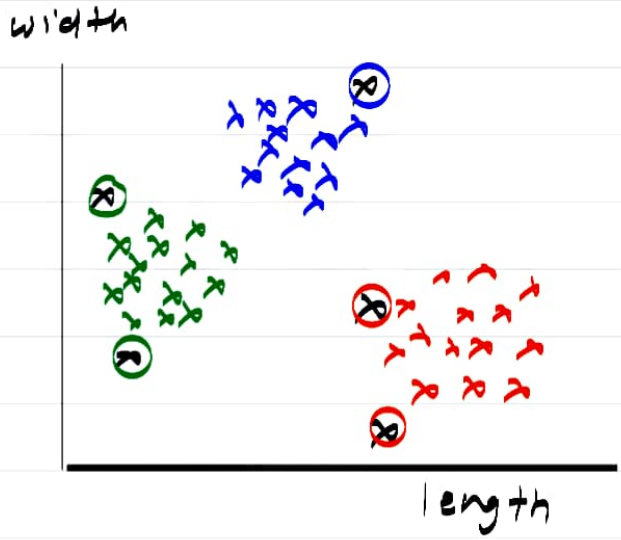
- In some cases, if a good assumption is not found then other option is to replace null values with mean/mode/median.
  (There are high chaces that this might not give good results)

We can also use <u>KNN method</u> for this.
K nearest neighbours

eg. two flowers data

| length | width | color | type |
|---|---|---|---|
| 10 | 2 | red | A |
| 8 | 2 | red | B |
| 7 | 4 | green | A |
| 5 | 3 | blue | A |
| 6 | 2 | ⋮ | B |
| 8 | 4 | ⋮ | ⋮ |
| 9 | 3 | ⋮ | ⋮ |
| ⋮ | ⋮ | | |
| null % | 0% | 0% | 25% | 0% |

apply KNN on these column



## 2) Outliers handling

- outliers confuse our ML model and affect its training

eg.

| area | rooms | price |
|---|---|---|
| 110 | 1 | 100 |
| 120 | 2 | 120 |
| 112 | 1 | 102 |
| 113 | 1 | 105 |
| 110 | 1 | 117 |
| 115 | 1 | 100 |
| 112 | 1 | 200 |
| 120 | 2 | 140 |
| 130 | 2 | 135 |
| 140 | 2 | 150 |

model will get confused due to this outlier

Outlier handling → 1) outlier detection
2) outlier handling

### 1) Outlier detection

detection using → iqr limits $Q_1 - 1.5\,iqr,\ Q_3 + 1.5\,iqr$
sigma limits $\mu \pm n\sigma$
↓
for continuous data

for categorical data we can use proportions to detect outliers

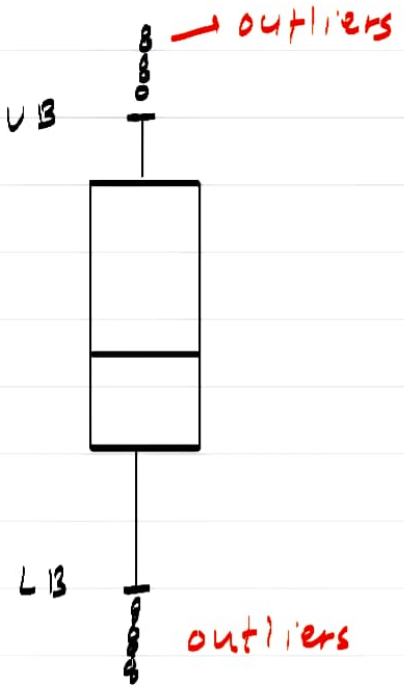
eg. categorical data

A → 70%

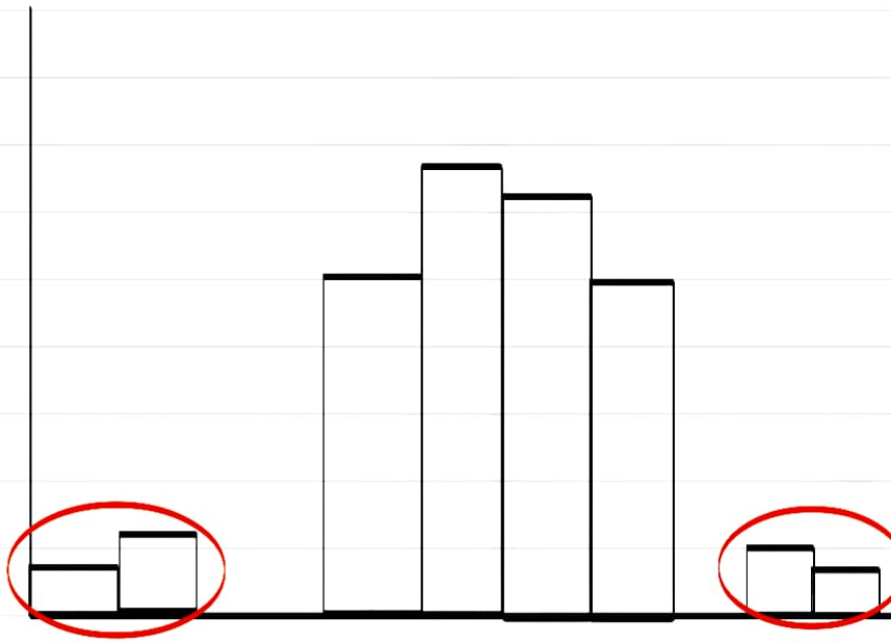
B → 28%

C → 2% → outlier category

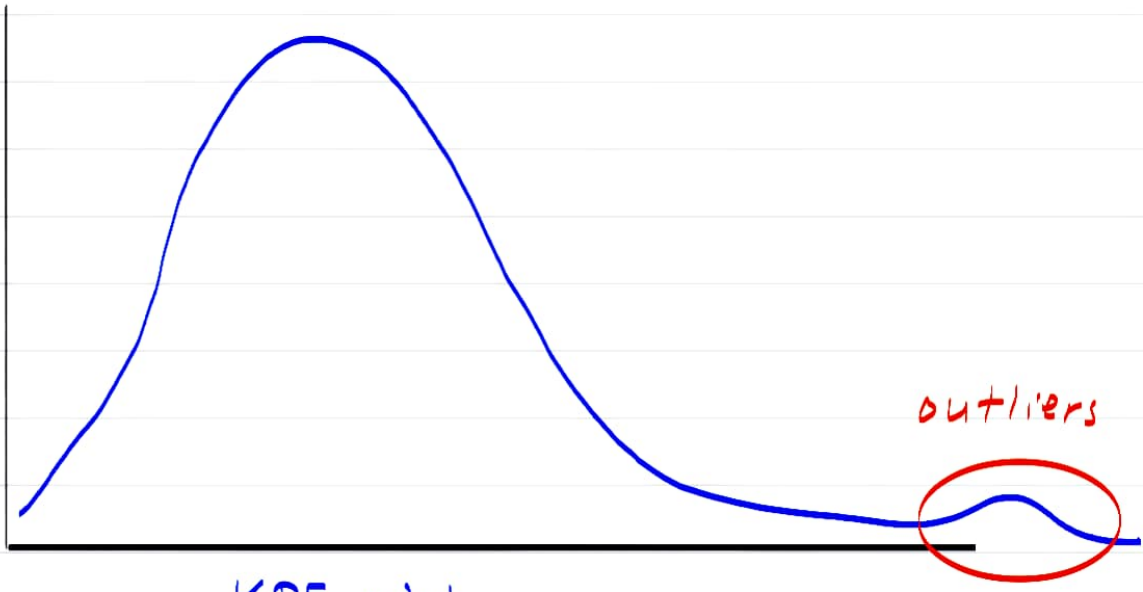
Visual methods of outlier detection




Box plot



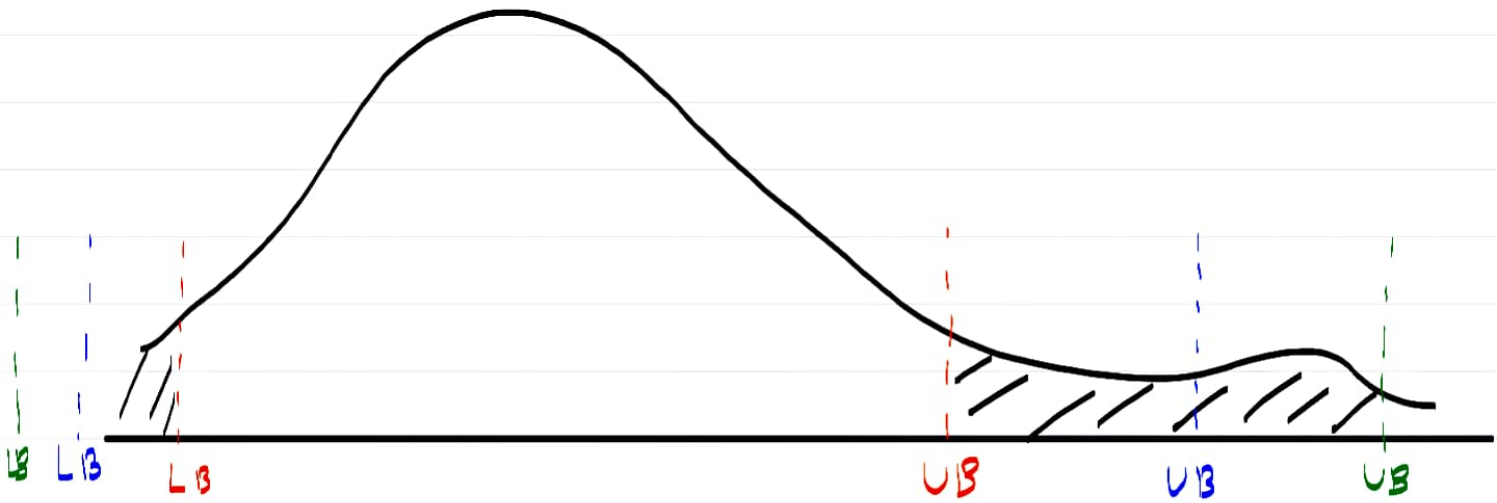
histogram




KDE plot

- After detection, we can remove outliers if
  - amount of outliers is very less eg. < 5%
    and
  - the outliers are not so important for given use case

if the % number of outliers is very large/very small then change the UB upper bound and LB lower bound values accordingly




if we can't remove the outliers (due to domain reasons) we need to do something to reduce impact of the outliers on analysis and ML model traini

solution - data transformation techniques like
- log. transformation ✓
  box-cox transformation
  feature engineering
  standardization